5 During cosmic-ray experiments in the 1930s, researchers observed the track of a previously unknown positively charged particle in a cloud chamber. The particle was measured to have a charge of +1 *e* and is known as the muon.

The particle enters a uniform magnetic field of flux density 0.250 T with a speed of $2.40 \times 10^{7}\ \text{m s}^{-1}$. Its velocity makes an angle of 60° with the direction of the magnetic field, causing it to follow a helical path as seen in Fig. 5.1 below.

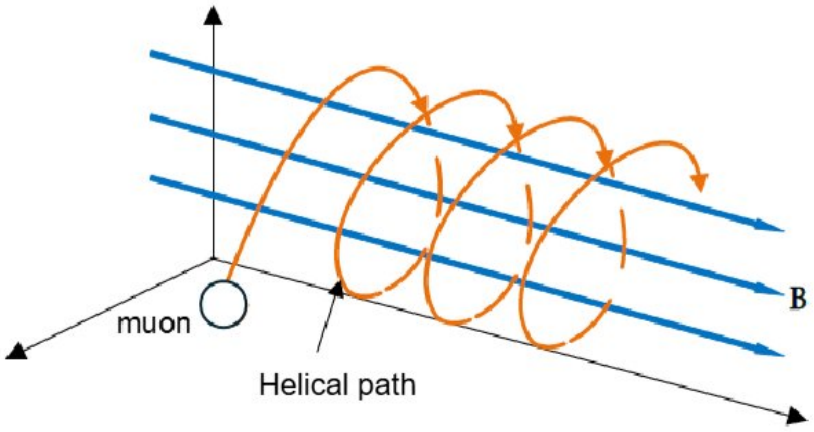


Fig. 5.1

Measurements of the track show that the muon took 2.96×10^{-8} s to complete one revolution.

(a) (i) Show that the radius of the helical path is 0.0979 m.

[2]

(ii) Determine the mass, m_μ of the muon.

m_μ = ... kg [2]

(b) Determine the distance travelled by the particle **parallel** to the magnetic field during one complete revolution.

distance = ... m [2]

(c) Before taking measurements, the researchers compensate for the vertical component of the Earth's magnetic field.

A short magnetic needle is suspended from a string 5.0 cm from a horizontal long straight wire, as shown in the top view in Fig. 5.2.

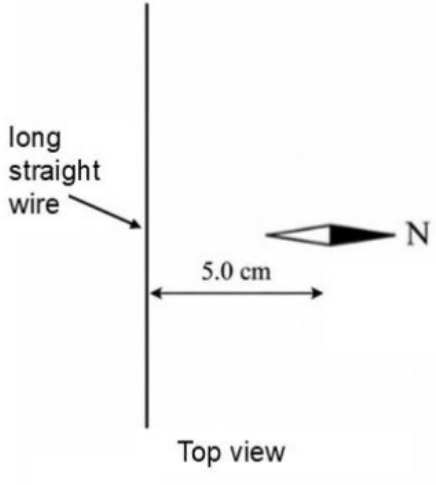


Fig. 5.2

When there is no current in the wire, the north pole of the needle dips at an angle of 6.0° below the horizontal, as shown in the side view in Fig. 5.3.

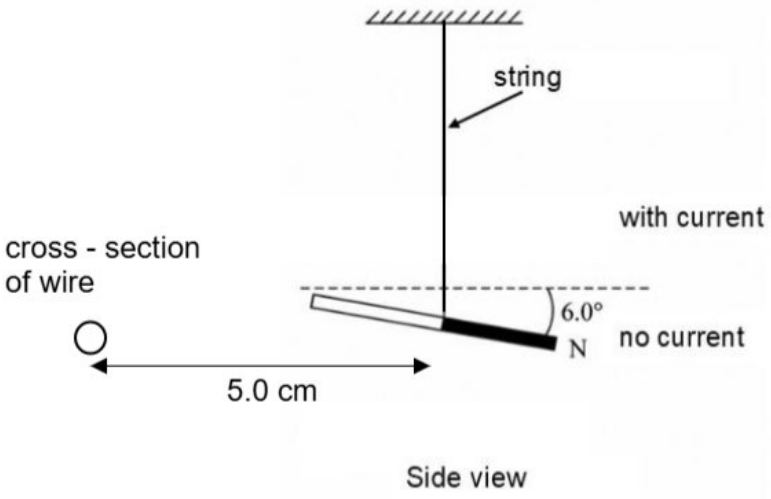


Fig. 5.3

A current of 0.47 A is then passed through the wire. The magnetic field produced by the current causes the needle to become horizontal, as shown in the side view in Fig. 5.4.

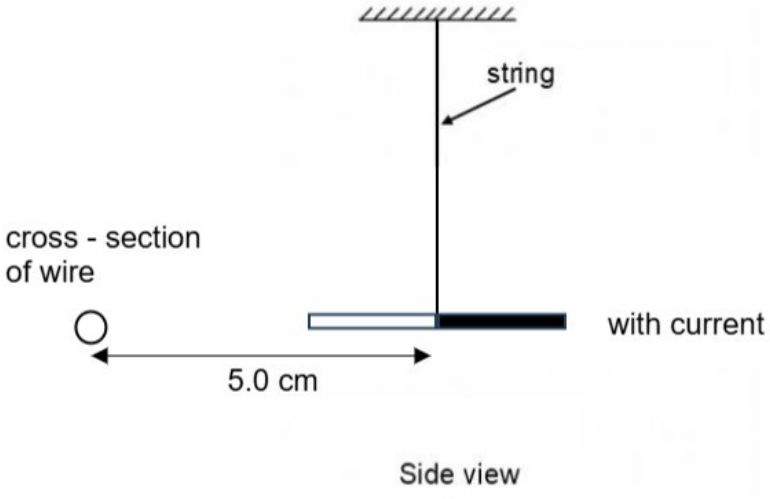


Fig. 5.4

(i) Show that the magnetic flux density produced by the wire, when the current is flowing through it, at the position of the short magnetic needle is 1.9×10^{-6} T.

[1]

(ii) Determine the magnitude of the Earth's magnetic flux density in the laboratory.

magnetic flux density = ...T [2]

Total: [9]